

Fernanda Costa<sup>1</sup>, Lucas Oliveira<sup>2</sup>, Mateus Silva<sup>2</sup>, and Camila Santos<sup>2</sup>

<sup>1</sup>University of Sao Paulo

<sup>2</sup>University of Sao

September 10, 2024

# End to end perception and prediction for Autonomous vehicles: A Survey

Fernanda Costa  
University of Sao Paulo, Brazil

Lucas Oliveira  
University of Sao Paulo, Brazil

Mateus Silva  
University of Sao Paulo, Brazil

Camila Santos  
University of Sao Paulo, Brazil

## Abstract

*The development of autonomous vehicles (AVs) has seen rapid advancements over the past decade, driven by significant improvements in artificial intelligence, sensor technologies, and computational power. This survey paper provides a comprehensive review of end-to-end perception and prediction systems for autonomous vehicles, which are critical components for enabling safe and efficient navigation in complex environments. We systematically examine the state-of-the-art approaches, focusing on deep learning-based methods that integrate perception (e.g., object detection, semantic segmentation) with prediction (e.g., trajectory forecasting) in a unified framework. The paper explores the key challenges associated with end-to-end learning, such as data fusion from multiple sensors, the trade-offs between accuracy and computational efficiency, and the interpretability of learned models. Additionally, we review publicly available datasets and benchmarks that have catalyzed progress in this domain. Finally, we discuss the current limitations and potential future research directions, highlighting the need for more robust, scalable, and generalizable solutions to achieve full autonomy in diverse driving scenarios. This survey aims to serve as a valuable resource for researchers and practitioners working towards the next generation of autonomous vehicle technologies.*

## 1. Introduction

The pursuit of fully autonomous vehicles (AVs) represents one of the most ambitious and transformative technological challenges of our time. Autonomous vehicles promise to revolutionize transportation, offering the potential for safer roads, reduced traffic congestion, and enhanced mobility for individuals unable to drive. As we edge closer to realizing this vision, significant strides have been made in the core technologies that enable autonomous driving, particularly

in the areas of perception and prediction.

Perception and prediction are fundamental to the operation of autonomous vehicles. Perception involves the ability of the vehicle to sense and understand its environment, identifying and classifying objects such as other vehicles, pedestrians, cyclists, and road signs. This process relies heavily on data from various sensors, including cameras, LiDAR, radar, and ultrasonic sensors. The integration and interpretation of this data are critical for constructing an accurate and reliable representation of the vehicle's surroundings.

Prediction, on the other hand, involves anticipating the future actions and trajectories of the perceived entities. This capability is essential for safe navigation and decision-making, allowing the vehicle to proactively adjust its path to avoid collisions and adhere to traffic rules. Effective prediction requires not only a deep understanding of the current scene but also the ability to model the intent and behavior of dynamic agents in the environment.

Historically, the perception and prediction tasks in autonomous vehicles were handled by separate, modular systems. However, this traditional approach often led to sub-optimal performance due to the complex interactions and dependencies between the two tasks. In recent years, the emergence of deep learning and end-to-end learning paradigms has catalyzed a shift towards more integrated approaches. These end-to-end systems aim to unify perception and prediction within a single, cohesive framework, leveraging the power of neural networks to directly map raw sensor inputs to actionable predictions.

End-to-end learning offers several advantages over traditional modular pipelines. By learning the entire process as a single model, these systems can optimize for global performance, potentially leading to more accurate and robust outcomes. Additionally, end-to-end approaches can reduce the need for hand-crafted features and heuristics, allowing the model to discover the most relevant patterns directly from the data. This holistic approach also facilitates better scala-

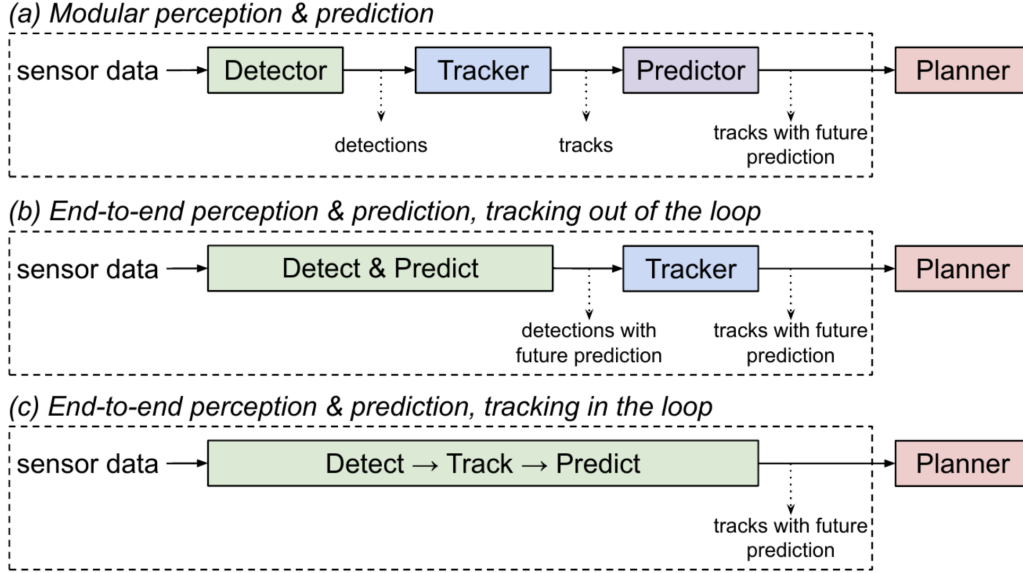


Figure 1. Different architectures for perception and prediction that are used in an autonomous vehicle.

bility and adaptability to different driving environments and conditions.

Despite these promising advancements, end-to-end perception and prediction systems face numerous challenges. The high-dimensionality and heterogeneity of sensor data, the need for real-time processing, and the interpretability of deep learning models are significant hurdles that must be addressed. Moreover, the safety-critical nature of autonomous driving demands an exceptionally high level of reliability and robustness, far beyond what is required in most other applications of artificial intelligence.

This survey aims to provide a comprehensive review of the current state of end-to-end perception and prediction for autonomous vehicles. We begin by outlining the foundational concepts and methodologies that underpin these systems, followed by a detailed examination of recent advances in deep learning techniques for perception and prediction. We then explore the integration of these components in end-to-end frameworks, discussing the benefits and limitations of this approach. Additionally, we review the key datasets and benchmarks that have driven progress in this field, and we analyze the performance of leading systems on these benchmarks.

Finally, we identify the key challenges and open research questions that remain in the journey toward fully autonomous driving. We highlight the need for further research in areas such as sensor fusion, model interpretability, and the handling of rare and edge-case scenarios. By providing a thorough overview of the current landscape, this survey seeks to guide researchers and practitioners in the development of more capable, reliable, and scalable au-

tonomous vehicle systems.

## 2. Top Approaches for End-to-End Perception and Prediction in Autonomous Vehicles

End-to-end perception and prediction systems for autonomous vehicles aim to directly map raw sensory inputs to driving decisions, integrating complex processes such as object detection, scene understanding, and motion forecasting into a unified model. Various approaches have been developed to tackle this problem, each with its own strengths, challenges, and applications. Below, we provide a detailed examination of the top approaches that have been employed in this domain. [1, 6, 27]

### 2.1. Convolutional Neural Networks (CNNs) and Variants

Convolutional Neural Networks (CNNs) have been at the forefront of computer vision tasks, making them a natural choice for perception in autonomous vehicles. CNNs excel at processing grid-like data, such as images, making them ideal for tasks like object detection, semantic segmentation, and scene understanding. [7, 9, 12, 16, 22, 24]

- **Traditional CNNs for Perception:** In an end-to-end framework, CNNs can be used to extract features from camera inputs, identifying objects, lanes, and other critical elements in the driving environment. These features can then be fed into prediction modules to forecast the future positions and trajectories of dynamic objects.
- **Region-Based CNNs (R-CNNs):** These networks have been adapted to focus on specific regions of interest within an image, which is particularly useful in au-

onomous driving for detecting and tracking multiple objects simultaneously.

- Fully Convolutional Networks (FCNs): FCNs are an extension of CNNs where all layers are convolutional, making them suitable for dense prediction tasks such as semantic segmentation. In an end-to-end system, FCNs can be used to generate pixel-level predictions that inform higher-level prediction tasks.
- Residual Networks (ResNets) and Variants: ResNets introduce skip connections to overcome the vanishing gradient problem, enabling the training of very deep networks. These networks can extract more complex and abstract features, improving the robustness of perception and prediction.

### 2.1.1 Challenges

- CNNs generally require a significant amount of labeled data for training.
- The computational cost of running deep CNNs in real-time on embedded systems can be high.
- They are typically focused on single-modality inputs, making it challenging to integrate multiple sensor data streams.

## 2.2. Recurrent Neural Networks (RNNs) and Long Short-Term Memory Networks (LSTMs)

RNNs, and their more sophisticated variants such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs), are particularly well-suited for sequence prediction tasks. In the context of autonomous driving, these networks are often used for temporal modeling and trajectory prediction. [17, 18]

- RNNs for Temporal Data: RNNs can process sequences of data, making them ideal for understanding how objects move over time. When integrated into an end-to-end system, RNNs can take the features extracted by CNNs (e.g., object positions) and predict their future states.
- LSTMs for Enhanced Prediction: LSTMs address the vanishing gradient problem in traditional RNNs by incorporating memory cells that can store information over long periods. This is crucial for predicting complex behaviors in dynamic environments, such as anticipating a pedestrian's intent to cross the street or a vehicle's lane-changing maneuver.
- Spatio-Temporal Networks: Combining CNNs with RNNs/LSTMs can lead to Spatio-temporal models that simultaneously process spatial information (e.g., object location in an image) and temporal information (e.g., object motion over time). This combination is powerful for tasks like multi-step trajectory prediction in autonomous vehicles.

### 2.2.1 Challenges

- Training RNNs, particularly LSTMs, can be computationally expensive and time-consuming.
- These models require a large amount of sequential data, which may not always be available.
- RNNs can struggle with very long sequences, which may lead to inaccurate predictions over extended periods.

## 2.3. Transformer Networks and Attention Mechanisms

Transformer networks, originally designed for natural language processing, have gained popularity in computer vision and sequential modeling tasks due to their ability to handle long-range dependencies using self-attention mechanisms. [3, 5, 8, 10, 13–15, 19, 20, 23, 25]

- Vision Transformers (ViTs): Vision Transformers have been adapted to process image data by treating images as sequences of patches. This approach allows for capturing global context more effectively than CNNs. In an end-to-end system, ViTs can be used to process visual data, providing a holistic understanding of the scene that informs prediction models.
- Attention Mechanisms for Prediction: Attention mechanisms allow the model to focus on the most relevant parts of the input when making predictions. This is particularly useful in dynamic driving environments where certain objects or regions of interest (e.g., an oncoming car) are more critical to the driving decision than others.
- Transformer-Based Sequence Modeling: Transformers can replace RNNs/LSTMs for sequence prediction, offering better scalability and parallelization. In autonomous driving, they can be used for predicting the future trajectories of multiple agents by effectively capturing the interactions between them.

### 2.3.1 Challenges

- Transformers require a large amount of data and computational power for training.
- They can be less intuitive and harder to interpret compared to traditional models like CNNs and RNNs.
- The integration of Transformers with multi-modal sensor data (e.g., LiDAR, radar) in end-to-end systems is still an emerging area of research.

## 2.4. Graph Neural Networks (GNNs)

Graph Neural Networks (GNNs) have recently gained attention for their ability to model relationships and interactions between different entities in a structured way. In the context of autonomous vehicles, GNNs are particularly useful for modeling the interactions between multiple agents (e.g., vehicles, pedestrians) in a driving environment.

- **GNNs for Interaction Modeling:** GNNs can represent each agent in a driving scenario as a node in a graph, with edges capturing the relationships or interactions between agents. This approach is powerful for tasks like multi-agent trajectory prediction, where the future state of each agent depends not only on its own motion but also on the behaviors of surrounding agents.
- **GNNs Integrated with CNNs/RNNs:** In end-to-end systems, GNNs can be combined with CNNs (for perception) and RNNs (for temporal modeling) to create a comprehensive model that processes visual data, understands temporal sequences, and predicts interactions among multiple agents.
- **Scene Graphs:** A specific application of GNNs is the construction of scene graphs, where objects and their spatial relationships are represented as a graph. This allows for a structured understanding of the scene that can improve prediction accuracy, particularly in complex environments.

#### 2.4.1 Challenges

- GNNs are relatively new in the field of autonomous driving, and their integration with other models is still being explored.
- They require careful design of graph structures and relationship modeling, which can be complex.
- The computational cost of GNNs can be high, especially for large-scale, real-time applications.

### 2.5. Reinforcement Learning (RL) and Imitation Learning (IL)

Reinforcement Learning (RL) and Imitation Learning (IL) offer frameworks for training autonomous driving models through interaction with the environment or by learning from demonstrations.

- **Reinforcement Learning:** In RL, an autonomous vehicle learns to make driving decisions by receiving rewards or penalties based on its actions. This approach is particularly suitable for learning complex driving policies in end-to-end systems, where the vehicle must navigate challenging scenarios such as avoiding obstacles, merging into traffic, or handling unprotected left turns.
- **Deep Reinforcement Learning (DRL):** DRL combines deep learning with reinforcement learning, allowing the model to process high-dimensional sensory inputs like images. In end-to-end systems, DRL can be used to train models that directly map raw sensor data to driving actions, optimizing for long-term safety and efficiency.
- **Imitation Learning:** IL involves training a model by mimicking expert behavior, typically through supervised learning. The model learns to predict actions based on observations, guided by a dataset of expert demonstrations.

In an end-to-end system, IL can be used to train the vehicle to replicate human driving behaviors, providing a baseline that can be further refined through RL or other techniques.

- **Hybrid RL/IL Approaches:** A combination of RL and IL can be employed, where the model first learns basic driving skills through imitation and then fine-tunes its performance through reinforcement learning. This hybrid approach can accelerate training and improve the robustness of the end-to-end system.

#### 2.5.1 Challenges

- RL requires extensive exploration, which can be risky and impractical in real-world driving scenarios, leading to a reliance on simulation environments.
- The training process for RL and DRL can be very slow and resource-intensive.
- IL models may struggle to generalize beyond the scenarios covered in the training data, making them less effective in novel or rare situations.

### 2.6. Multi-Modal Fusion Models

Autonomous vehicles rely on a variety of sensors, including cameras, LiDAR, radar, and GPS, each providing complementary information about the driving environment. Multi-modal fusion models aim to integrate these diverse data streams into a cohesive representation that can be used for perception and prediction.

- **Early Fusion:** In early fusion approaches, data from different sensors are combined at the raw data level (e.g., concatenating RGB images with LiDAR point clouds) before being processed by a neural network. This approach can capture the correlations between different sensor modalities from the outset.
- **Late Fusion:** Late fusion involves processing each sensor's data independently through separate networks and then combining the resulting features at a later stage. This approach allows each sensor modality to be processed optimally according to its specific characteristics before integration.
- **Intermediate Fusion:** Intermediate fusion, or hybrid fusion, combines elements of both early and late fusion. It allows for partial integration of sensor data at multiple stages of the processing pipeline, leveraging the strengths of both approaches.

#### 2.6.1 Challenges

- Fusion models can be complex and computationally expensive, particularly in real-time

### 3. Datasets for End-to-End Perception and Prediction in Autonomous Vehicles

The development and evaluation of end-to-end perception and prediction systems for autonomous vehicles heavily rely on high-quality datasets. These datasets provide the necessary data to train deep learning models and benchmark their performance in real-world scenarios. A robust dataset should include a diverse set of driving conditions, environments, and object types, along with accurately labeled ground truth data for both perception and prediction tasks. Below, we discuss several key datasets that are widely used in the research community for developing and testing end-to-end autonomous driving systems. [2, 4, 11, 21, 26]

#### 3.1. KITTI Dataset

The KITTI Vision Benchmark Suite is one of the most renowned datasets in the autonomous driving research community. Collected by the Karlsruhe Institute of Technology and Toyota Technological Institute at Chicago, the KITTI dataset includes data captured from a variety of sensors mounted on a car, including cameras, LiDAR, and GPS/IMU. The dataset covers urban, rural, and highway driving scenarios, with annotations for object detection, tracking, and semantic segmentation. Although primarily used for perception tasks, the dataset also supports research in prediction through its tracking and scene flow annotations.

##### 3.1.1 Advantages

- Extensive sensor suite including stereo cameras and 3D LiDAR.
- High-quality annotations for a wide range of perception tasks.
- Diverse driving scenarios that cover different environments and traffic conditions.

##### 3.1.2 Limitations

- Limited size compared to more recent datasets.
- Focuses more on perception tasks with fewer resources dedicated to prediction.

#### 3.2. nuScenes Dataset

The nuScenes dataset, created by Motional, offers a large-scale, richly annotated dataset specifically designed for autonomous driving research. It includes data from six cameras, a 32-beam LiDAR, five radars, and GPS/IMU sensors, covering 1,000 scenes collected in Boston and Singapore. The dataset provides annotations for object detection, tracking, and attribute prediction, as well as detailed information about the environment, including weather and time of day.

##### 3.2.1 Advantages

- Comprehensive sensor suite and a large number of annotated scenes.
- Annotations include object attributes and tracking, supporting both perception and prediction tasks.
- Coverage of complex urban environments with challenging traffic conditions.

##### 3.2.2 Limitations

- High complexity in data processing and annotation due to the large size and multi-sensor nature.

#### 3.3. Waymo Open Dataset

The Waymo Open Dataset is one of the largest publicly available datasets for autonomous driving. It includes data collected from five high-resolution LiDARs and five cameras mounted on vehicles driving in diverse environments, primarily in the U.S. The dataset provides annotations for 3D object detection, tracking, and prediction, making it suitable for end-to-end perception and prediction research.

##### 3.3.1 Advantages

- Large-scale dataset with millions of annotated objects.
- Rich sensor data including high-resolution LiDAR and cameras.
- Comprehensive annotations for 3D detection, tracking, and prediction.

##### 3.3.2 Limitations

- Primarily U.S.-based, which might limit generalizability to different geographic regions.
- Dataset size poses challenges for storage and processing.

#### 3.4. Argoverse Dataset

The Argoverse dataset, developed by Argo AI, is specifically designed for autonomous vehicle research, with a strong focus on motion forecasting and prediction. It includes two primary components: Argoverse 3D Tracking, which provides annotated LiDAR and camera data for object detection and tracking, and Argoverse Motion Forecasting, which includes over 300,000 scenarios with rich annotations for vehicle trajectories. This makes it particularly valuable for research in end-to-end systems that integrate perception and prediction.

##### 3.4.1 Advantages

- Focus on motion forecasting with extensive trajectory data.
- Diverse urban driving scenarios.

- Well-structured dataset with clear annotations for prediction tasks.

### 3.4.2 Limitations

- Limited geographic diversity as the data is primarily collected in Pittsburgh, Miami, and other U.S. cities.
- Smaller in size compared to other large-scale datasets like Waymo.

## 3.5. ApolloScape Dataset

The ApolloScape dataset, provided by Baidu's Apollo platform, offers a large collection of images, 3D point clouds, and annotations for various autonomous driving tasks. The dataset includes challenging driving scenarios in diverse environments, including highways, city streets, and residential areas. Annotations cover semantic segmentation, instance segmentation, lane marking, and object detection, with the additional capability for trajectory prediction.

### 3.5.1 Advantages

- Wide range of driving environments and conditions.
- Rich annotations supporting both perception and prediction tasks.
- Availability of challenging scenarios like night driving and complex intersections.

### 3.5.2 Limitations

- Primarily focuses on the Chinese driving environment, which may differ from Western datasets.
- Requires significant computational resources for processing the large data volumes.

## 4. Summary

The field of autonomous driving has seen remarkable progress in recent years, with end-to-end perception and prediction systems emerging as a promising approach to achieving full vehicle autonomy. These systems offer the potential to streamline the complex pipeline of traditional modular systems, providing a more integrated and efficient means of processing sensor data and making driving decisions. By leveraging advanced deep learning techniques, such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Transformer Networks, Graph Neural Networks (GNNs), and Reinforcement Learning (RL), researchers and practitioners are pushing the boundaries of what autonomous vehicles can achieve in diverse and dynamic environments.

However, the journey towards fully autonomous vehicles is far from complete. Each of the top approaches discussed in this paper presents unique strengths and challenges, from

the high computational demands of deep neural networks to the complexity of integrating multiple sensor modalities. While end-to-end systems have demonstrated significant promise in simulation environments and controlled real-world scenarios, achieving the level of robustness, reliability, and safety required for deployment in everyday traffic remains a critical challenge.

Moreover, the reliance on large-scale, high-quality datasets underscores the importance of continued data collection and annotation efforts to capture the full spectrum of driving scenarios. As the field advances, there will be a growing need for more sophisticated datasets that not only provide richer sensory inputs but also represent a broader range of driving environments, including rare and edge-case scenarios that are critical for safe autonomous operation.

Future research must also address key issues such as the interpretability of end-to-end models, the fusion of heterogeneous sensor data, and the ability to generalize across different geographical regions and driving conditions. The development of hybrid approaches that combine the strengths of various deep learning models, along with advances in real-time processing capabilities, will be essential for overcoming these challenges.

In conclusion, while significant progress has been made, the development of end-to-end perception and prediction systems for autonomous vehicles is still an active and rapidly evolving area of research. By continuing to refine these approaches and addressing their current limitations, the goal of achieving fully autonomous vehicles that can safely navigate the complexities of the real world moves closer to reality. This paper has aimed to provide a comprehensive overview of the state-of-the-art in this field, serving as a resource for researchers and practitioners as they contribute to the next generation of autonomous driving technologies.

## References

- [1] Alex Bewley, Zongyuan Ge, Lionel Ott, Fabio Ramos, and Ben Upcroft. Simple online and realtime tracking. In *2016 IEEE international conference on image processing (ICIP)*, pages 3464–3468. IEEE, 2016. 2
- [2] Holger Caesar, Varun Bankiti, Alex H Lang, Sourabh Vora, Venice Erin Liong, Qiang Xu, Anush Krishnan, Yu Pan, Giancarlo Baldan, and Oscar Beijbom. nuscenes: A multi-modal dataset for autonomous driving. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11621–11631, 2020. 5
- [3] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. In *European conference on computer vision*, pages 213–229. Springer, 2020. 3
- [4] Ming-Fang Chang, John Lambert, Patsorn Sangkloy, Jagjeet Singh, Slawomir Bak, Andrew Hartnett, De Wang, Peter Carr, Simon Lucey, Deva Ramanan, et al. Argoverse: 3d tracking and forecasting with rich maps. In *Proceedings of*



the *IEEE/CVF conference on computer vision and pattern recognition*, pages 8748–8757, 2019. 5

- [5] Alexey Dosovitskiy. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 3
- [6] Yunhao Du, Zhicheng Zhao, Yang Song, Yanyun Zhao, Fei Su, Tao Gong, and Hongying Meng. Strongsort: Make deepsort great again. *IEEE Transactions on Multimedia*, 25: 8725–8737, 2023. 2
- [7] Kaiwen Duan, Song Bai, Lingxi Xie, Honggang Qi, Qingming Huang, and Qi Tian. Centernet: Keypoint triplets for object detection. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 6569–6578, 2019. 2
- [8] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4015–4026, 2023. 3
- [9] Hei Law and Jia Deng. Cornernet: Detecting objects as paired keypoints. In *Proceedings of the European conference on computer vision (ECCV)*, pages 734–750, 2018. 2
- [10] Zhiqi Li, Wenhai Wang, Hongyang Li, Enze Xie, Chonghao Sima, Tong Lu, Yu Qiao, and Jifeng Dai. Bevformer: Learning bird’s-eye-view representation from multi-camera images via spatiotemporal transformers. In *European conference on computer vision*, pages 1–18. Springer, 2022. 3
- [11] Yiyi Liao, Jun Xie, and Andreas Geiger. Kitti-360: A novel dataset and benchmarks for urban scene understanding in 2d and 3d. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(3):3292–3310, 2022. 5
- [12] Wei Liu, Dragomir Anguelov, Dumitru Erhan, Christian Szegedy, Scott Reed, Cheng-Yang Fu, and Alexander C Berg. Ssd: Single shot multibox detector. In *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part I 14*, pages 21–37. Springer, 2016. 2
- [13] Yingfei Liu, Tiancai Wang, Xiangyu Zhang, and Jian Sun. Petr: Position embedding transformation for multi-view 3d object detection. In *European Conference on Computer Vision*, pages 531–548. Springer, 2022. 3
- [14] Yingfei Liu, Junjie Yan, Fan Jia, Shuailin Li, Aqi Gao, Tiancai Wang, and Xiangyu Zhang. Petrv2: A unified framework for 3d perception from multi-camera images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 3262–3272, 2023.
- [15] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. 3
- [16] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 779–788, 2016. 2
- [17] Robin M Schmidt. Recurrent neural networks (rnns): A gentle introduction and overview. *arXiv preprint arXiv:1912.05911*, 2019. 3
- [18] Alex Sherstinsky. Fundamentals of recurrent neural network (rnn) and long short-term memory (lstm) network. *Physica D: Nonlinear Phenomena*, 404:132306, 2020. 3
- [19] Apoorv Singh. Vision-radar fusion for robotics bev detections: A survey. In *2023 IEEE Intelligent Vehicles Symposium (IV)*, pages 1–7, 2023. 3
- [20] Apoorv Singh. Surround-view vision-based 3d detection for autonomous driving: A survey. In *2023 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, pages 3235–3244, 2023. 3
- [21] Pei Sun, Henrik Kretzschmar, Xerxes Dotiwalla, Aurelien Chouard, Vijaysai Patnaik, Paul Tsui, James Guo, Yin Zhou, Yuning Chai, Benjamin Caine, et al. Scalability in perception for autonomous driving: Waymo open dataset. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 2446–2454, 2020. 5
- [22] Zhi Tian, Xiangxiang Chu, Xiaoming Wang, Xiaolin Wei, and Chunhua Shen. Fully convolutional one-stage 3d object detection on lidar range images. *Advances in Neural Information Processing Systems*, 35:34899–34911, 2022. 2
- [23] A Vaswani. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017. 3
- [24] Tai Wang, Xinge Zhu, Jiangmiao Pang, and Dahua Lin. Fcos3d: Fully convolutional one-stage monocular 3d object detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 913–922, 2021. 2
- [25] Yue Wang, Vitor Campagnolo Guizilini, Tianyuan Zhang, Yilun Wang, Hang Zhao, and Justin Solomon. Detr3d: 3d object detection from multi-view images via 3d-to-2d queries. In *Conference on Robot Learning*, pages 180–191. PMLR, 2022. 3
- [26] Benjamin Wilson, William Qi, Tanmay Agarwal, John Lambert, Jagjeet Singh, Siddhesh Khandelwal, Bowen Pan, Ratnesh Kumar, Andrew Hartnett, Jhony Kaesemodel Pontes, et al. Argoverse 2: Next generation datasets for self-driving perception and forecasting. *arXiv preprint arXiv:2301.00493*, 2023. 5
- [27] Nicolai Wojke, Alex Bewley, and Dietrich Paulus. Simple online and realtime tracking with a deep association metric. In *2017 IEEE international conference on image processing (ICIP)*, pages 3645–3649. IEEE, 2017. 2